

Causal Inference



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THE REMIX

Advanced Topics in Applied Causal Inference

Day 1 of 5: Foundations

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This week

Mon May 18	Foundations — the 2×2 , parallel trends, event studies
Tue May 19	Covariates and staggered adoption
Wed May 20	Synthetic control and the frontier
Thu May 21	TBD
Fri May 22	TBD

Who am I?

Scott Cunningham · Professor of Economics, Baylor University

- Empirical work: sex work, relationship markets, drug policy, abortion, corrections, suicide and mental illness — and AI agents and AI methodologies
- Podcast: *The Mixtape with Scott*
- Substack: *Scott's Substack*
- Teaching platform: *Mixtape Sessions*

We'll show Claude Code for research throughout the week

- Live demonstrations: building, debugging, and reasoning with an AI agent in your terminal
- Three days of casual integration — not a Claude Code course, but an honest look at the workflow
- Bring your laptop if you want to follow along

If you've never tried it, that's fine. Watch first; install later.

AI agents touch research in two places

1. Production

Agents that *write* the research — pose questions, pull data, run analyses, draft papers.

2. Verification

Agents that *audit* the research — check identification, robustness, hidden choices, replicability.

Both are growing fast. Both matter for what you do.

And yet — my audits find surprising variation

- Highly scaffolded, narrowly restricted use of agents on DiD
- Across runs: **variation in specifications, sample restrictions, and conclusions**
- Signs of specification searching — even when agents are warned about known biased estimators
- The production layer is not transparent. You have to audit it.

Competence in *both* technique and interpretation

- **Technique:** the estimators, the assumptions, the conditions under which they hold
- **Interpretation:** what the number means, when it's wrong, what alternative designs would say
- Neither one alone is enough — and agents are not yet able to do either reliably without you watching

Returns to econometric skill are *higher*, not lower

- As we rely more on APE, we'll be less hands-on with the code
- That's exactly why **continuing education and workshops matter more, not less**
- Ongoing investment in your own skill becomes the differentiator
- **That's why we're spending three days on something as basic as DiD.** You have to know it cold.

Pass / fail: a dataset, an analysis, an interpretation

- Seven students writing for credit
- I'll provide a dataset and a research question
- You'll run an analysis using the methods we cover and write a short interpretation
- Not real-research scale — enough to test that you can apply the design

Format finalizing with Dan; details by Wednesday.

Today's seven lessons

1. Three founders of DiD: Semmelweis, Snow, Ashenfelter
2. Three regressions, one answer
3. Potential outcomes, target parameters, and parallel trends
4. Standard errors: vanilla, robust, cluster-robust
5. The minimum wage wars
6. Event studies as falsifications — and 2×2 s
7. Two ATTs, two parallel trends — a callback to weighting

LESSON

1. DiD Origins

DiD is a research design, not a regression



A research design is an identification strategy: it reveals the invisible target parameter hidden inside a calculation.

DiD has two ingredients



Together they identify the ATT.

One of several causal panel methods

- **Difference-in-differences** — today and tomorrow
- Synthetic control — Wednesday
- Panel IV, matrix completion, factor models — mentioned, not deep-dived
- Each is a different research design. Different assumptions. Different target parameters.

DiD has three inventors

- **Ignaz Semmelweis** — Vienna, 1847. Two clinics, one intervention, a clean comparison.
- **John Snow** — London, 1854. Two water companies, a natural experiment.
- **Orley Ashenfelter** — Princeton, 1970s. Formalizes the calculation: “four averages and three subtractions.”

Each one ran a DiD before the name existed. We'll meet them in order.

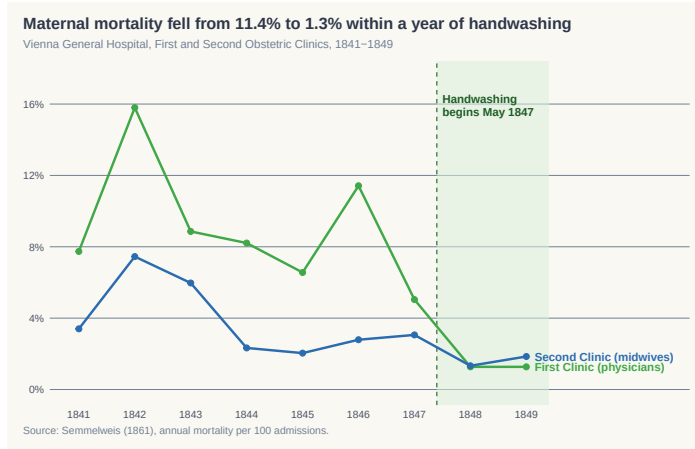
Vienna, 1840s: two clinics, two mortality rates

- Vienna General Hospital, First Obstetric Clinic
- Pregnant women were assigned to the *physicians' wing* or the *midwives' wing* by day of admission — effectively at random
- Physicians' wing: **13–18% mortality** from puerperal fever
- Midwives' wing: **about 3%**
- Women would deliver in the street rather than risk the physicians' wing

His hypothesis, and his DiD

- Semmelweis — an assistant on the wards — noticed the physicians did cadaver dissections before delivering babies. The midwives did not.
- Hypothesis: something on the physicians' hands was causing the fever.
- May 1847: he instituted **chlorinated lime handwashing for physicians only**
- Midwives served as the untreated comparison group — a DiD by hand

The data



Monthly mortality, First Obstetric Clinic. Handwashing began mid-1847.

Rejected

- Superiors attributed the drop to a new ventilation system
- Miasma theory of disease was still dominant
- Semmelweis was passed over for promotion, eventually left Vienna
- Published his major treatise in 1861; largely ignored
- Committed to an asylum in 1865, died there at 47 — from infected wounds, the very condition he had been fighting

Pasteur and Lister vindicated germ theory in the years that followed.

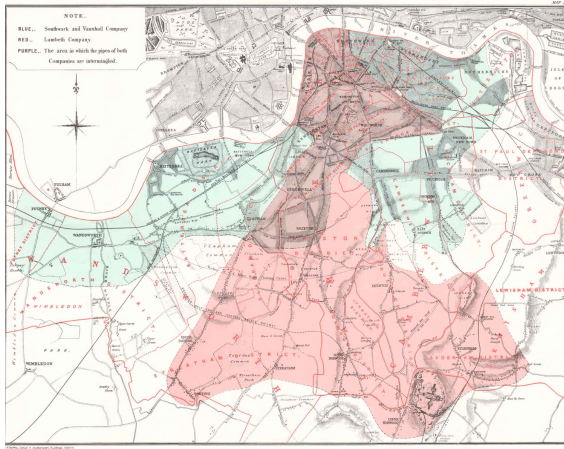
London, 1854: John Snow vs. miasma

- Three cholera waves had hit London since the 1830s
- Dominant theory: miasma — “foul air”
- Snow’s theory: a waterborne pathogen, ingested, expelled, recirculated
- In 1854, an outbreak gave him a natural experiment

The Metropolis Water Act of 1852

- Required London water companies to move their intake pipes upstream, above the city sewage
- **Lambeth** moved its intake to Thames Ditton in 1852 — clean water
- **Southwark & Vauxhall** delayed — still drew from polluted central Thames
- Both served overlapping London neighborhoods. Same streets, different water.

Customers of two companies, the same streets



Snow walked the streets, recording each household's water company.

Snow's DiD — two paths to the same number

	1849 outbreak <i>(pre-treatment)</i>	1854 outbreak <i>(post-treatment)</i>	Change → <i>(over time)</i>
Lambeth (treated)	$\bar{Y}_{LB}$	$\bar{Y}_{LA}$	$\bar{Y}_{LA} - \bar{Y}_{LB}$
Southwark & Vauxhall (control)	$\bar{Y}_{SB}$	$\bar{Y}_{SA}$	$\bar{Y}_{SA} - \bar{Y}_{SB}$
Change ↓ <i>(between companies)</i>	$\bar{Y}_{LB} - \bar{Y}_{SB}$	$\bar{Y}_{LA} - \bar{Y}_{SA}$	DiD

$$\widehat{\text{DiD}} = \underbrace{(\bar{Y}_{LA} - \bar{Y}_{LB}) - (\bar{Y}_{SA} - \bar{Y}_{SB})}_{\text{right, then down}} = \underbrace{(\bar{Y}_{LA} - \bar{Y}_{SA}) - (\bar{Y}_{LB} - \bar{Y}_{SB})}_{\text{down, then right}}$$

Same four cells. Same answer. Under PT, that number is the ATT for the houses Lambeth supplied.

The same two paths are two regressions

RIGHT, THEN DOWN

First-difference each unit, then regress on the treatment dummy.

1. $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$
2. $\Delta Y_i = \alpha + \delta D_i + u_i$

Slope $\hat{\delta}$ is the DiD.

DOWN, THEN RIGHT

Compute the within-period treated-vs-control gap, then regress on Post.

1. $G_t = \bar{Y}_{t,\text{treat}} - \bar{Y}_{t,\text{control}}$
2. $G_t = \alpha + \delta \text{Post}_t + u_t$

Slope $\hat{\delta}$ is the same DiD.

```
# Path 1 -- right then down: unit first-differences on D
wide <- df |> pivot_wider(names_from = period, values_from = Y)
wide$d <- wide$post - wide$pre
coef(lm(d ~ treat, data = wide))["treat"] # = DiD

# Path 2 -- down then right: period-level gap on Post
gap <- df |> group_by(period) |>
  summarize(G = mean(Y[treat == 1]) - mean(Y[treat == 0]))
gap$post <- c(0, 1)
coef(lm(G ~ post, data = gap))["post"] # = same DiD
```

Snow identified cholera *before* germ theory

- Cholera mortality in Lambeth-supplied houses: a small fraction of Southwark & Vauxhall houses
- The water hypothesis was right — but the pathogen wouldn't be isolated for another **29 years** (Koch, 1883)
- Snow died in 1858, also before vindication
- This is identification by design, decades before the term existed

Princeton's Industrial Relations Section

The Section was older than Princeton's economics department — founded 1922, with a non-partisan mission: **rigorous empirical “manpower” studies**. Worker mobility, unions, wages, federal training programs — the policy edge of US labor economics.

Orley Ashenfelter's leadership (1970s onward)

- Long-time director. Set the tone: high-quality data, careful design, communicate to bureaucrats.
- Trained a remarkable cohort of students through the Section:
David Card, Josh Angrist, Bob LaLonde, Janet Currie, plus Steve Pischke, John Dinardo, Anne Case, and many others.
- Hired **Alan Krueger** as a junior colleague from Harvard — not a student, but a teammate.

A lot of what we now call “the credibility revolution” came out of this one floor in Firestone Library.

The empirical crisis that gave us DiD

By the late 1970s, empirical labor economics was in trouble. Different methods on the same question *disagreed wildly*. Three pieces crystallized it:

- **LaLonde (1986)** ran the standard methods on a job-training RCT (NSW, $ATE \approx +\$800$) after replacing the experimental control with survey data. Couldn't recover the truth. Methods disagreed by orders of magnitude.
- **Heckman** (*Handbook of Labor Economics*, female labor supply): estimates “all over the place” — many probably weren't credible to begin with.
- **Ashenfelter & Card (1985, ReStat)**: even DiD wasn't an obvious winner on job training. Honest about its limits from the start.

The Princeton response: design over modeling

The IRS pushed for estimators that **mimic an RCT as closely as possible**. DiD was born inside this push — a step back toward design.

Hear Card on the crisis (26:20): youtu.be/1soLdywFb_Q?t=1580

Orley's answer: *keep it simple*

- Princeton, Industrial Relations Section, late 1970s.
- Worker surveys — matching trainees in federal job-training programs to non-trainees.
- Massive panel regressions: *individual* fixed effects + *time* fixed effects.
- Audience: federal labor-policy bureaucrats who held the budget.

Their eyes would glaze over whenever you said ‘regression.’

— Orley Ashenfelter, in conversation with Scott

*The constraint shaped the method. **DiD was meant to be simple** — communicable, defensible, transparent. Not the most sophisticated estimator on the menu. The most useful one.*

His move: “four averages and three subtractions”

- Stop leading with the regression
- Lead with the thing the regression is computing — the **four cell means** and the **three subtractions** between them
- Then, if anyone wants the equation, give them the equation

Regressions are average-making machines anyway.

Hear Orley coin the term

- **Orley coins “difference-in-differences”** so he doesn't have to explain regression to bureaucrats.

youtu.be/WnB3EJ8K7lg (3:53 long)

Worth hearing in his own voice. The phrase was a rhetorical choice before it was an estimator name.

The four averages

	Pre ($t = 0$)	Post ($t = 1$)
Control	$\bar{Y}_{D=0, t=0}$	$\bar{Y}_{D=0, t=1}$
Treated	$\bar{Y}_{D=1, t=0}$	$\bar{Y}_{D=1, t=1}$

Four cells. Four averages.

The three subtractions

$$\widehat{\text{DiD}} = \underbrace{(\bar{Y}_{11} - \bar{Y}_{10})}_{\text{treated change}} - \underbrace{(\bar{Y}_{01} - \bar{Y}_{00})}_{\text{control change}}$$

Three subtractions because the outer minus is the third.

castle.dta: stand-your-ground laws, U.S. states

- Cheng & Hoekstra (2013): did expanding self-defense laws raise homicide rates?
- State-year panel
- Today: restrict to **2005–2006** — a clean 2×2
- Treated states: those that adopted in 2006. Control: those that hadn't.

Loads directly: use

<https://github.com/scunning1975/mixtape/raw/master/castle.dta>

The four cells, by hand — in Stata

```
use castle.dta, clear
keep if year >= 2005 & year <= 2006
gen post = (year == 2006)
gen treat = (effyear == 2006)

* Four cell means
summarize l_homicide if treat==1 & post==1 // Y11
summarize l_homicide if treat==1 & post==0 // Y10
summarize l_homicide if treat==0 & post==1 // Y01
summarize l_homicide if treat==0 & post==0 // Y00
```

The four cells, by hand — in R

```
library(haven); library(dplyr)
df <- read_dta("castle.dta") |>
  filter(year %in% 2005:2006) |>
  mutate(post = as.integer(year == 2006),
         treat = as.integer(effyear == 2006))

df |>
  group_by(treat, post) |>
  summarize(Ybar = mean(l_homicide), .groups = "drop")
```

The four cells, by hand — in Python

```
import pandas as pd
df = pd.read_stata("castle.dta")
df = df.query("year >= 2005 & year <= 2006").copy()
df["post"] = (df.year == 2006).astype(int)
df["treat"] = (df.effyear == 2006).astype(int)

df.groupby(["treat", "post"])["l_homicide"].mean()
```


We get a number. Hold it.

$$\widehat{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$$

This is the answer. Now we'll get it three more ways.

Same data. Three languages. One number.

Stata

`display (y11 - y10) - (y01 - y00)` $\hat{\delta} \approx \mathbf{0.10}$

R

`(Y11 - Y10) - (Y01 - Y00)` $\hat{\delta} \approx \mathbf{0.10}$

Python

`(y11 - y10) - (y01 - y00)` $\hat{\delta} \approx \mathbf{0.10}$

*Log homicide rate is about 10% higher in 2006-adopting states relative to non-adopting states.
Stand-your-ground in real numbers.*

LESSON

2. Three Regressions, One Answer

Three regressions, three languages

- I'll show the **Stata** version for each of the three specifications
- The full multi-language versions are in:
 - `decks/code/equivalence/equivalence.do`
 - `decks/code/equivalence/equivalence.R`
 - `decks/code/equivalence/equivalence.py`
- All three languages, all three regressions, the same number to machine epsilon

Regression 1: OLS with interactions

```
reg l_homicide post##treat, cluster(sid)
```

The coefficient on `1.post#1.treat` is the DiD.

The saturated specification: $Y = \beta_0 + \beta_1 \text{Post} + \beta_2 \text{Treat} + \beta_3 (\text{Post} \times \text{Treat}) + \varepsilon$

Regression 2: Two-way fixed effects

```
xtset sid year  
xtreg l_homicide c.treat#c.post i.year, fe vce(cluster sid)
```

State fixed effects absorb time-invariant differences. Year fixed effects absorb common shocks.

*In the 2×2 case, the TWFE coefficient equals the DiD.
Tomorrow we'll see when that equivalence breaks.*

Regression 3: Long differences

```
keep sid year l_homicide treat  
reshape wide l_homicide, i(sid) j(year)  
gen diff = l_homicide2006 - l_homicide2005  
reg diff treat, vce(cluster sid)
```

Collapse to one observation per state. The slope on treat is the DiD.

Time-differencing eliminates the unit fixed effects mechanically.

All three return the same number

Regression 1 (OLS with interactions)	$\hat{\beta}_3$
Regression 2 (TWFE)	$\hat{\beta}_3$
Regression 3 (Long differences)	$\hat{\beta}_3$
Four averages, three subtractions	$\hat{\beta}_3$

Identical to machine epsilon.

This equivalence is what made DiD extensible

- OLS infrastructure absorbs covariates, fixed effects, multiple groups, multiple periods
- The 2×2 generalizes through the regression
- **But** the equivalence breaks once you add complexity
- OLS then needs *additional assumptions* for the coefficients to retain a causal interpretation

Tomorrow: covariates, staggered adoption, continuous treatment. Each one breaks something.

LESSON

3. Potential Outcomes,

Target Parameters, and Parallel Trends

Ingredient 1: done. Ingredient 2: now.



The calculation gives us a number. The assumption tells us what the number means.

Princeton econometrics, 1980s

- Joshua Angrist's recollection: in Princeton's PhD program, potential outcomes were not taught
- Econometrics was about *coefficients* — “the structure,” “the causal effect” — not about weighted averages of unit-level effects
- The vocabulary that lets us write $Y_i(1) - Y_i(0)$ was **not yet standard in economics**

Rubin (1974) revives Neyman (1923)

- Jerzy Neyman, 1923 dissertation: introduced potential outcomes notation
- Donald Rubin, 1974: “Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies”
- Together they gave us the explicit language of **counterfactuals** — $Y_i(1)$ and $Y_i(0)$ as the two states of the same unit

50-year gap. The notation existed before economists started using it.

The individual treatment effect

$$\delta_i = Y_i(1) - Y_i(0)$$

- One per unit. **Different** across units in general — treatment effects are heterogeneous.
- We only ever see one of $Y_i(1)$ or $Y_i(0)$. The other is the counterfactual.
- So δ_i is never directly observed at the individual level.

Three averages of δ_i — three estimands

All three are expected values. The only difference is *whom we average over*:

- **ATE** = $\mathbb{E}[\delta_i]$ *average over everyone*
- **ATT** = $\mathbb{E}[\delta_i \mid D_i = 1]$ *average over treated units only*
- **ATU** = $\mathbb{E}[\delta_i \mid D_i = 0]$ *average over untreated units only*

$$\text{ATE} = \pi \cdot \text{ATT} + (1 - \pi) \cdot \text{ATU}, \quad \text{where } \pi = \Pr(D = 1).$$

With heterogeneous effects, the three are different numbers. The one you want depends on the question.

Eight people, three averages

i	$Y_i(0)$	$Y_i(1)$	δ_i	D_i	Y_i (observed)
1	5	9	4	1	9
2	4	10	6	1	10
3	6	11	5	1	11
4	7	12	5	1	12
5	5	8	3	1	8
6	2	3	1	0	2
7	3	5	2	0	3
8	1	4	3	0	1

$$\mathbf{ATE} = \frac{1}{8}(4+6+5+5+3+1+2+3) = \mathbf{3.625} \quad \mathbf{ATT} = \frac{1}{5}(4+6+5+5+3) = \mathbf{4.6} \quad \mathbf{ATU} = \frac{1}{3}(1+2+3) = \mathbf{2.0}$$

Three different numbers. If we had both potential outcomes, this is the arithmetic.

Which causal effect do you want? Weighting decides.

Solon, Haider & Wooldridge (2015), *What Are We Weighting For?*

- In a survey, you weight to make estimates **nationally representative**.
- In causal inference, you weight because you want a **different parameter**.
- Different question. Different reason.

ATE, ATT, ATU are all expected values. But which expectation? Over which units? With what weights? That's not a survey-design question — it's a parameter-of-interest question.

Ten people, two counties

Person	$Y(0)$	$Y(1)$	$\delta_i = Y(1) - Y(0)$	County
Alan	0	1	1	1
Betty	1	1	0	1
Chad	1	1	0	1
Daniel	0	0	0	1
Edith	0	1	1	1
Frank	0	1	1	1
George	0	0	0	1
Hank	0	1	1	1
Ida	1	0	-1	2
Janet	1	0	-1	2

County 1: eight people. County 2: two people. Compute the average δ for each county — and the average δ across all ten. Do you get the same answer?

Same data, two weights, two parameters

- **Within counties:** County 1 avg $\delta = \frac{1}{8}(1+0+0+0+1+1+0+1) = +0.5$.
County 2 avg $\delta = \frac{1}{2}(-1+(-1)) = -1.0$.
- **ATE for the average person:** $\frac{1}{10}(8 \cdot 0.5 + 2 \cdot (-1)) = +0.2$
- **ATE for the average county:** $\frac{1}{2}(0.5 + (-1)) = -0.25$

*Same potential outcomes. **Different signs.** Whichever you report depends on what you're trying to answer.*

Scale it up: 254 counties, 30 million people

You just saw it with 10 people. Now blow it up.

Texas: 254 counties. The 5 biggest (Harris, Dallas, Tarrant, Bexar, Travis) hold 41% of the population. The other 249 share the rest.

The setup

Every person has their own treatment effect δ_i .

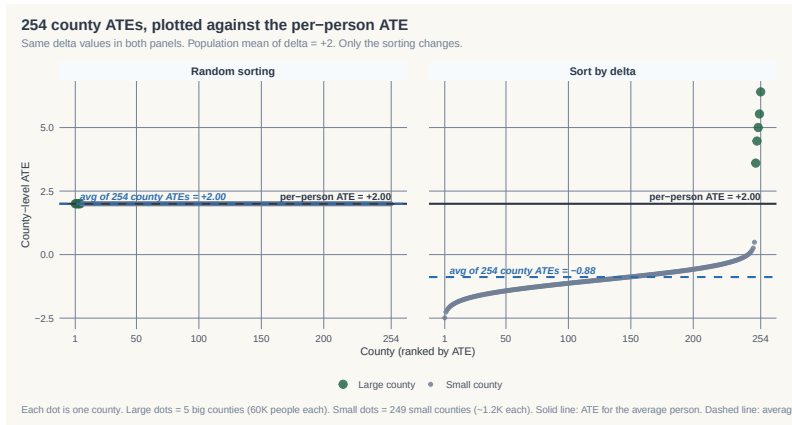
Half: $\delta_i \sim \mathcal{N}(+5, 1)$. Half: $\delta_i \sim \mathcal{N}(-1, 0.5)$. $\mathbb{E}[\delta] = +2$.

We assign people to counties *two* different ways:

- ▶ **Random** — each county is a random sample of the population.
- ▶ **Sort by δ** — high-effect people end up in the 5 big counties.

The δ_i values are identical in both scenarios. Only who-lives-where changes.

Sorting on δ can flip the sign of your answer



Random: every county's ATE equals the population's. δ -sort: per-person line at +2, avg of county ATEs flips to -0.88 .

Random sorting is the benchmark — rarely reality. People choose where to live, and on traits that move δ .

Two parameters, two answers

Under δ -sort, the 5 big counties cluster at $\approx +5$. The 249 small counties cluster at ≈ -1 .

- **Per-person ATE:** $\frac{1}{N} \sum_i \delta_i$ — weight every person by $1/N$.
The 15M high- δ people balance the 15M low- δ people. **Answer: +2.**
- **Avg of county ATEs:** $\frac{1}{254} \sum_c \bar{\delta}_c$ — weight every county by $1/254$.
 $\frac{5(+5)+249(-1)}{254} = \textbf{-0.88}$. The 249 small counties dominate.

Same data. Two weights. Two parameters. Sometimes two signs.

What this means for DiD on aggregate data

DiD usually runs on aggregates: counties, states, hospitals.

- Equal weights per unit $\Rightarrow$ **unit-weighted ATE**: the effect for the average *location*.
- Weight by population $\Rightarrow$ **person-weighted ATE**: the effect for the average *person*.

Which one you want depends on the question

Person-weighted for **aggregate human welfare**: how many people are affected, by how much?

Unit-weighted when locations have **autonomy and goals**: each policy unit chooses independently — the unit *is* the agent.

Solon, Haider & Wooldridge (2015). Code: `decks/code/Tiebout-Roy/tiebout_roy_chart.R`

The weighted mean — a single formula

$$\bar{Y}_w = \frac{\sum_i w_i Y_i}{\sum_i w_i}$$

The weight is the choice:

- $w_i = 1$ for every unit $\Rightarrow$ **unit-weighted** (the average county).
- $w_i = \text{population}$ $\Rightarrow$ **person-weighted** (the average person).

Two weights, two signs

Same 10 people. County 1 DiD = +0.5 (8 people). County 2 DiD = -1.0 (2 people).

Person-weighted:
$$\frac{8 \cdot (+0.5) + 2 \cdot (-1.0)}{8 + 2} = +0.2$$

Unit-weighted:
$$\frac{(+0.5) + (-1.0)}{2} = -0.25$$

Same data. Different weight. Different sign.

Castle data: the weighted 2×2 by hand

Four cell means, each with [aw=popwt]:

```
summarize l_homicide if treat==1 & post==1 [aw=popwt]
gen wy11 = 'r(mean)'
summarize l_homicide if treat==1 & post==0 [aw=popwt]
gen wy10 = 'r(mean)'
summarize l_homicide if treat==0 & post==1 [aw=popwt]
gen wy01 = 'r(mean)'
summarize l_homicide if treat==0 & post==0 [aw=popwt]
gen wy00 = 'r(mean)'

gen wdid = (wy11 - wy10) - (wy01 - wy00)
display wdid // 0.0964
```

Same number via regression: Stata

```
reg l_homicide post##treat [aweight=popwt], cluster(sid)
```

Coefficient on 1.post#1.treat = **0.0964**

aweight is the formula — run inside OLS.

Same number via regression: R

```
library(fixest)

feols(l_homicide ~ post * treat,
      data = castle,
      weights = ~popwt,
      cluster = ~sid)
```

Coefficient on `post:treat` = **0.0964**

Same number via regression: Python

```
import statsmodels.formula.api as smf

mod = smf.wls("l_homicide ~ post * treat",
              data = castle,
              weights = castle["popwt"]).fit(
    cov_type = "cluster",
    cov_kws = {"groups": castle["sid"]})
```

`mod.params["post:treat"] = 0.0964`

Pick a target parameter

Everything until now has been about *parameters*: ATE, ATT, ATU.

- Three weights. Three signs. Three answers.
- The choice is about *which causal effect you actually want*.

Today's target: the ATT.

The average effect for the units that actually received the treatment.

The switching equation: you see one half

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

- For each unit: $Y_i(1)$ if treated, $Y_i(0)$ if not.
- ATT asks about *both* columns. The data hands you one.

The rest of the day: strategies for recovering the ATT from the half you see.

What if we just compared Y between treated and untreated?

The simple difference in observed outcomes (SDO):

$$\text{SDO} = \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]$$

For our 8 people:

$$\text{SDO} = \frac{1}{5}(9+10+11+12+8) - \frac{1}{3}(2+3+1) = 10 - 2 = 8$$

But $\text{ATT} = 4.6$. $\text{ATE} = 3.625$. $\text{ATU} = 2.0$.

The SDO is bigger than any of them.

$$\text{SDO} = \text{ATT} + \text{selection bias}$$

Add and subtract $\mathbb{E}[Y_i(0) \mid D_i = 1]$ inside the SDO and rearrange:

$$\underbrace{\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]}_{\text{SDO}} = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid D = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]}_{\text{selection bias}}$$

In the 8-person example:

$$\text{selection bias} = \frac{1}{5}(5+4+6+7+5) - \frac{1}{3}(2+3+1) = 5.4 - 2.0 = \mathbf{3.4}$$

$$\text{SDO} = 4.6 + 3.4 = 8 \quad \checkmark$$

Calculations alone aren't enough

- The SDO is something we can *always* calculate from the data.
- It is the population estimand of the naive comparison.
- But it is **not the target parameter** — in general, $SDO \neq ATT \neq ATE \neq ATU$.
- The gap is the selection bias. It does not go away by collecting more data.

To turn a calculation into a causal target, we need an *identification strategy*

RCT: random assignment makes selection bias zero.

IV: relevance + exclusion.

RDD: continuity at the cutoff.

DiD: parallel trends + no anticipation + stability (SUTVA).

Let's connect causal inference with statistics for a second

- So far the language has been *causal*: $Y(1)$, $Y(0)$, the switching equation, the counterfactual.
- Causal claims are about **populations** — the average effect on the treated, $\mathbb{E}[Y(1) - Y(0) \mid D = 1]$.
- But we never see the population. We see a sample.
- Which means we need the *statistical* layer too: sampling distributions, estimators, convergence.

The two layers interlock — one tells us what we want to estimate; the other tells us how close our estimate gets.

Suppose you had all the data

What would you compute?

*A regression. A difference of means. A ratio.
Whatever the data permits.*

That's a *population estimand*

Population estimand

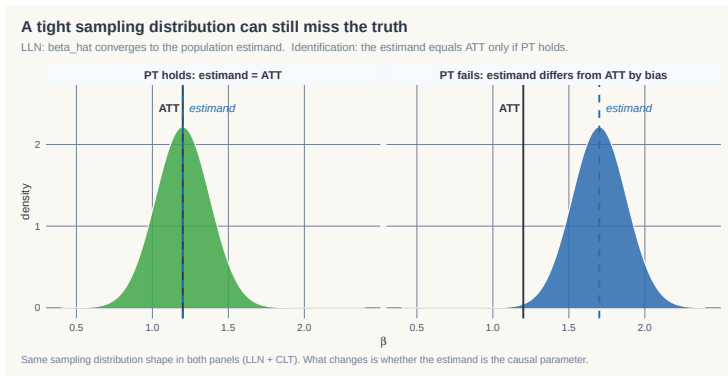
The calculation you could do if you had the entire population.

A function of population moments — means, variances, regressions.

Not a sample quantity. **Not** necessarily causal.

Sometimes denoted θ . The thing your estimator is converging to in the limit.

Shape is statistics. Center is identification.



Shape of the sampling distribution $\leftarrow$ LLN + CLT. Independent of whether the design is causal.

Center of the sampling distribution $\leftarrow$ identification. Equals ATT only if PT holds.

Same data, same estimator, same statistics — two completely different conclusions.

Regress earnings on a college dummy. Causal?

$$\beta = \mathbb{E}[\text{Earnings} \mid \text{College} = 1] - \mathbb{E}[\text{Earnings} \mid \text{College} = 0]$$

Is β the causal effect of college?

Yes if randomized. No if selection.

Randomized

Treatment assigned by coin flip.
Treated and untreated populations are identical in expectation.

β is the causal effect.

Selection

College-goers differ from non-college-goers in ability, family background, ambition.

β is biased.

An estimand can be biased. A target parameter cannot.

Population estimand

What the math gives you with all the data.

Sometimes equals the causal effect. Sometimes doesn't.

Target parameter

The causal object you actually want.

Defined in potential-outcomes notation.

Unbiased by construction — it *is* the causal effect.

Bias is what rides along with the causal term

$$\underbrace{\text{Estimand}}_{\text{what the math gives}} = \underbrace{\text{Causal term}}_{\text{target parameter}} + \underbrace{\text{Bias}}_{\text{everything else}}$$

Bias is the gap. Sometimes zero (randomization). Often not.

Identification strips the bias

- Identification = a set of assumptions under which the bias term vanishes
- With those assumptions, the population estimand **equals** the target parameter
- Different designs use different identifying assumptions:
 - Randomization → ATE
 - **Parallel trends** → **ATT**
 - Instrumental variables → LATE
 - Regression discontinuity → local ATE

The ATT is the DiD target parameter

$$\text{ATT} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1, t = \text{Post}]$$

The average treatment effect on the treated, after treatment.

Treated units. Post-treatment period. Not the ATE.

The proof: add a zero, rearrange (five steps)

1. **Start with the 2×2 calculation** (four cell means, three subtractions):

$$\widehat{\text{DiD}} = [\mathbb{E}[Y \mid D=1, \text{Post}] - \mathbb{E}[Y \mid D=1, \text{Pre}]] - [\mathbb{E}[Y \mid D=0, \text{Post}] - \mathbb{E}[Y \mid D=0, \text{Pre}]]$$

2. **Apply the switching equation** $Y = D Y(1) + (1-D) Y(0)$. Treated-post is $Y(1)$; the other three cells are $Y(0)$:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]]$$

3. **Add a zero.** Insert $+\mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}]$ (the missing counterfactual for the treated group):

$$\begin{aligned} \widehat{\text{DiD}} &= \mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}] \\ &\quad + \mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]] \end{aligned}$$

4. **Rearrange.** First two terms = ATT; rest = a difference of differences in $Y(0)$ trends:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid 1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta \mathbb{E}[Y(0) \mid 1] - \Delta \mathbb{E}[Y(0) \mid 0]\}}_{\text{non-parallel-trends bias}}$$

5. **Read.** If the bias equals zero (*parallel trends*), then $\widehat{\text{DiD}} = \text{ATT}$.

The bias term *is* selection bias — on trends, not levels

The “non-parallel-trends bias” isn’t a new concept. It’s selection bias, in a different costume.

$$\text{bias} = \underbrace{\Delta \mathbb{E}[Y(0) \mid D=1]}_{\text{how treated's } Y(0) \text{ moves}} - \underbrace{\Delta \mathbb{E}[Y(0) \mid D=0]}_{\text{how controls' } Y(0) \text{ moves}}$$

- **Cross-section selection bias** asks: are treated and control different in *levels* of $Y(0)$?
- **DiD’s selection bias** asks: are treated and control different in *trends* of $Y(0)$?
- Different levels are fine. Unit fixed effects soak them up.
- What *can’t* differ is the *rate* at which $Y(0)$ moves across groups. If it does, the DiD reports the gap as treatment effect.

The trade DiD makes vs. randomization

RCTs: treated and controls identical in everything — levels and trends.

DiD: treated and controls only need to be identical in *trends* of $Y(0)$. **Cheaper assumption. Weaker identification.**

Four ways the 2×2 can go wrong

The proof on the last slide identifies the ATT **only** when three things hold:

- (i) **No Anticipation (NA)** — pre-period Y for the treated group equals $Y(0)$
- (ii) **Parallel Trends (PT)** — $Y(0)$ moves the same way in both groups
- (iii) The comparison group is itself **untreated**

Four scenarios. The same **+0 trick** works on all of them — only the zero you add changes.

- **Case 1.** NA ✓, never-treated control ✓

$$\widehat{\text{DiD}} = \text{ATT}$$

- **Case 2.** NA fails, never-treated control

$$\widehat{\text{DiD}} = \text{ATT} - \text{ATT}_{\text{pre}}$$

- **Case 3.** NA ✓, but control absorbs **spillover**

$$\widehat{\text{DiD}} = \text{ATT} - \text{spillover}$$

- **Case 4.** NA ✓, but control is **already-treated**

$$\widehat{\text{DiD}} = \text{ATT} - \Delta\text{ATT}$$

Each case: the +0 trick, a numerical example, a Monte Carlo verdict at the end.

Case 1. NA holds, never-treated comparison

Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$.

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(0)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Pre}]]$$

Add the zero. Insert $+\mathbb{E}[Y(0)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Post}]$. Rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta\mathbb{E}[Y(0)|1] - \Delta\mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}}$$

Numerical example

$$\text{ATT} = 10, \text{ PT bias} = 0. \quad \widehat{\text{DiD}} = 10. \quad \checkmark$$

Case 2. NA fails — anticipation contaminates Y_{Pre}

Suppose the treated group anticipates and adjusts *before* the policy lands. Then their pre-period outcome contains some $Y(1)$ response: $\mathbb{E}[Y|1, \text{Pre}] = \mathbb{E}[Y(0)|1, \text{Pre}] + \text{ATT}_{\text{pre}}$.

Same +0 trick, propagated:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}_{\text{Post}}}_{\text{what we want}} + \underbrace{\text{non-PT bias}}_{=0} - \underbrace{\text{ATT}_{\text{pre}}}_{\text{anticipation}}$$

Numerical example

$\text{ATT} = 10$, $\text{PT bias} = 0$, $\text{ATT}_{\text{pre}} = 3$. $\widehat{\text{DiD}} = 10 - 3 = 7$. **biased down**

If treatment effects are roughly constant, DiD can collapse to zero.

Case 3. The control group also receives treatment in Post

The comparison group's Post outcome is also $Y(1)$, not $Y(0)$. Switching equation:

$$\widehat{\text{DiD}} = \left[\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}] \right] - \left[\mathbb{E}[Y(1)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Pre}] \right]$$

Add two zeros — one missing counterfactual on each side:

$$+ \mathbb{E}[Y(0)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Post}] \quad \text{and} \quad + \mathbb{E}[Y(0)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Post}]$$

Rearrange. Treated side $\rightarrow$ ATT + treated $Y(0)$ trend. Control side $\rightarrow$ a *control-group* ATT + control $Y(0)$ trend.

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1)-Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\left\{ \Delta \mathbb{E}[Y(0)|1] - \Delta \mathbb{E}[Y(0)|0] \right\}}_{\text{non-PT bias} = 0 \text{ under PT}} - \underbrace{\mathbb{E}[Y(1)-Y(0)|0, \text{Post}]}_{\text{ATT}_{D=0, \text{Post}} \text{ (the spillover)}}$$

Numerical example

ATT = 10, PT bias = 0, $\text{ATT}_{D=0, \text{Post}} = 4$. $\widehat{\text{DiD}} = 10 - 4 = 6$. **biased toward zero**

Case 3, in shorthand

The decomposition from the previous slide, written compactly:

$$\widehat{\text{DiD}} = (+1) \cdot \text{ATT}_{\text{Post}} + (+1) \cdot \text{PTA} + (-1) \cdot \text{ATT}_{\text{Post}, D=0}$$

The weights sum to one: $+1 + 1 - 1 = 1$. But one weight is **negative**.

- A clean weighted average of ATTs has *all positive* weights.
- This isn't that. The spillover ATT enters with a -1 .
- Under PT (bias = 0), DiD is the treated ATT *minus* the control ATT.

Memorize the formula. Work the algebra. Watch for negative weights — they show up again in staggered DiD.

Case 4. The forbidden 2×2 — already-treated comparison

Now the “comparison” group was treated *before* the pre-period. Both their pre and post cells are $Y(1)$, with effects that can move over time: $\Delta ATT \equiv ATT(\text{Post}, D=0) - ATT(\text{Pre}, D=0)$.

Same recipe, three zeros (one per $Y(0)$ counterfactual we need):

$$\widehat{\text{DiD}} = \underbrace{ATT}_{\text{what we want}} + \underbrace{\text{non-PT bias}}_{=0} - \underbrace{\Delta ATT}_{\text{control's effect dynamics}}$$

Two numerical examples. $ATT = 10$, $\text{PT bias} = 0$:

- Control's $ATT(\text{Pre}) = 1$, $ATT(\text{Post}) = 5$, $\Delta ATT = 4$: $\widehat{\text{DiD}} = 10 - 4 = 6$
- Control's $ATT(\text{Pre}) = 1$, $ATT(\text{Post}) = 15$, $\Delta ATT = 14$: $\widehat{\text{DiD}} = 10 - 14 = -4$ **sign flip**

The Bacon insight

The forbidden 2×2 identifies $-\Delta ATT$. If treatment effects **grow over time**, DiD reports the *wrong sign*. Foreshadows Bacon (2021), tomorrow.

The Monte Carlo, in one R script

Where it lives:

```
decks/code/case_decompositions.R
```

What it does:

- Generates treatment effects for every unit (true $ATT = 10$).
- Simulates four DGPs — one per case from the last four slides.
- Repeats 2000 times. Records $\widehat{DiD}$ each time.
- Plots the sampling distribution of $\widehat{DiD}$ for each case.

If the algebra was right, the simulation lands where the algebra said.

Monte Carlo verdict — 2000 reps, four cases, one truth

Four ways the 2x2 can go wrong

Sampling distribution of $\hat{\beta}_{2 \times 2}$ over 2000 Monte Carlo reps · $n = 200$ per group · True ATT = 10

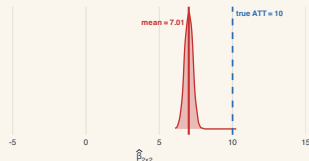
Case 1. NA holds, never-treated control

Predicted: $\beta = \text{ATT} = 10$



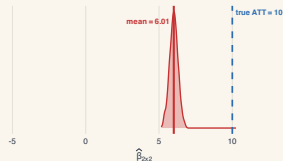
Case 2. NA fails (anticipation = 3)

Predicted: $\beta = \text{ATT} - \text{ATT}_{\text{pre}} = 10 - 3 = 7$



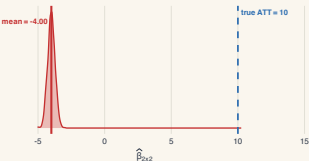
Case 3. Spillover into control (= 4)

Predicted: $\beta = \text{ATT} - \text{spillover} = 10 - 4 = 6$



Case 4. Already-treated control ($\Delta\text{ATT} = 14$)

Predicted: $\beta = \text{ATT} - \Delta\text{ATT} = 10 - 14 = -4$



Only Case 1 puts the truth at the center of the sampling distribution. The other three are sharp, narrow, and wrong.

No Anticipation — the other assumption

Step 2 of the proof used $Y_{\text{Pre}} = Y(0)$ for the treated group. That step has a name: **No Anticipation (NA)**.

What NA says

For any unit, $Y_t = Y_t(0)$ at every t before its treatment date. Units don't respond to a treatment they haven't received yet.

Holds trivially if *either*:

1. **Treatment effects are zero in the pre-period** — if a unit hasn't been treated, by construction $Y = Y(0)$.
2. **Treatment is unanticipated** — agents don't see it coming, so they can't react ahead of time.

Examples.

- *NA holds*: a state will let you drive without a license next year. You can't today. $Y = Y(0)$. ✓
- *NA fails*: you'll win the lottery tomorrow and you know it. You buy a house today. $Y \neq Y(0)$. ✗

The cost of losing No Anticipation

If the pre-period is contaminated by anticipation, Y_{Pre} for the treated group is $Y(1)$, not $Y(0)$. The 2×2 now identifies:

$$\widehat{DiD} = \underbrace{ATT_{Post}}_{\text{what we want}} + \underbrace{\text{non-PT bias}}_{\text{still there}} - \underbrace{ATT_{Pre}}_{\text{the anticipation effect}}$$

- If treatment effects are roughly constant over time, DiD can collapse to **zero** despite a real effect.
- Sign and magnitude of the bias depend on *when* you set the baseline.

The fix: roll the baseline back

Push the pre-period earlier — before any anticipation could plausibly have started. Clean $Y(0)$ baseline.

Side effect: PT now has to hold over the longer window. ATT is still identified.

Parallel trends: a different *kind* of assumption

Randomization is a design. The parent process *guarantees* full distributional equivalence between treatment groups. In the population, $\mathbb{E}[Y(1)|D=0] = \mathbb{E}[Y(1)|D=1]$. Plug-ins identify the ATE.

Parallel trends is a model. An asserted $Y(0)$ trend. Sibling to RDD smoothness and synthetic control's factor model. **No parent process backs it.**

Stronger? Weaker? Different kind of thing.

- **Weaker:** you only need equal $Y(0)$ *trends*, not full distributional equivalence.
- **Stronger:** you have to defend the assertion with no design that enforces it.

Pre-trends and falsifications are heuristics, not identification — they're finger-in-the-air checks for whether the asserted model looks reasonable.

The model-based family of identification

Three siblings. Each asserts something about $Y(0)$ that the data cannot directly confirm.

- **Parallel trends** (DiD): $\mathbb{E}[Y(0) \mid D=1]$ *trends* like $\mathbb{E}[Y(0) \mid D=0]$ in the post period.
- **Smoothness** (RDD): $\mathbb{E}[Y(0) \mid X=x]$ is continuous as x crosses the threshold.
- **Factor model** (synthetic control): $Y(0)_{it}$ follows a low-rank structure the donor pool spans.

None of these are designed. You assert them, then check that the asserted shape is consistent with what you can observe. The assumption itself lives in the counterfactual — where the data is silent.

What backs a model-based assumption

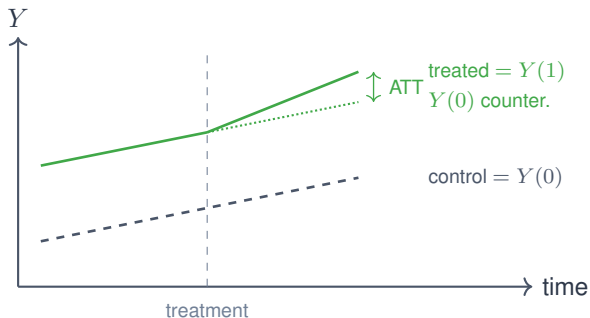
Randomization is its own defense. The lottery was fair; the assumption holds by construction.

A model-based assumption has no such defense. What backs it instead:

- **Institutional reasoning.** Why would treated and control move together absent treatment? What forces shaped both? Where does the variation in treatment come from?
- **Choice of comparison group.** Closer in covariates, exposure, and history → more credible.
- **Pre-period anchoring.** A long stable pre-period of parallel movement makes the asserted post-period trend easier to believe.

The most credible DiDs (Card-Krueger, Mariel) earn their credibility from the institutional story — not from the pre-trend test.

Parallel trends, drawn



*All three $Y(0)$ trajectories (control, treated-pre, counterfactual) share one slope.
Levels can differ. Trends must move together.*

Now we have both ingredients



Open the sheet — stability and trends, by hand

- Four tabs in sequence: Potential Outcomes $\rightarrow$ 2X2 $\rightarrow$ Balancing $\rightarrow$ 2XT #1
- Show: $Y = Y(0)$ pre-treatment, for everyone
- Show: $Y = Y(0)$ for control units, *regardless* of what treated units do
- That second point is **stability** — one unit's treatment doesn't change another's $Y(0)$

Sheet: <https://docs.google.com/spreadsheets/d/1onabpc14JdrGo6NFv0zCWo-nuWDLV2L1qNogDT9SBw/edit>

LESSON

4. Standard Errors

Vanilla, Robust, Cluster-Robust

DiD has two kinds of bias — not one

How Much Should We Trust Differences-in-Differences Estimates?

— Bertrand, Duflo & Mullainathan, *QJE* 2004

Two distinct failure modes

(1) **Biased estimator.** Non-parallel trends contaminate $\hat{\beta}$ — covered in Lessons 1–3.

(2) **Biased inference.** Even with $\hat{\beta}$ unbiased, a wrong SE gives the wrong t , p , and CI.

- Heteroskedasticity and within-cluster correlation are *inference* problems, not identification problems.
- BDM's alarm bell: DiD with vanilla SE rejects a true null at $\approx 45\%$ instead of 5%.
- This lesson: **get the spread right** so the test you reported is the test you actually ran.

The stakes: a wrong standard error is a false positive

We design hypothesis tests to control the false-positive rate:

$$\text{size} = \Pr(\text{reject } H_0 \mid H_0 \text{ is true}) \stackrel{!}{=} \alpha \text{ (typically 5\%).}$$

- Get the SE right $\Rightarrow$ you reject a true null $\approx 5\%$ of the time. As advertised.
- Get the SE *too small* $\Rightarrow |t|$ inflated $\Rightarrow$ you reject at **10%**, **20%**, sometimes **45%**.
You publish “significant” results that are noise.

The whole point of this lesson

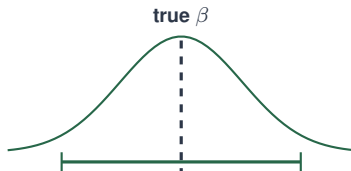
The OLS point estimate $\hat{\beta}$ is *fine* — unbiased for the ATT **under parallel trends**.

What we're fixing is the spread we report around it. Same data, same $\hat{\beta}$ — different inference.

Same $\hat{\beta}$, two reported spreads, two test sizes

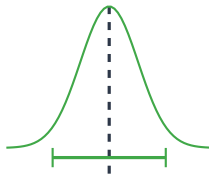
Correct SE

size $\approx 5\%$



Wrong SE

size $\gg 5\%$



$\hat{\beta}$ is in the same place — unbiasedness intact. The reported spread is wrong — and that IS the problem.
“You think you’re running a 5% test. You’re actually running a 15% test.”

The point estimate can be unbiased — the standard error often is not

- OLS gives us a $\hat{\beta}$ that converges to the right thing under PT + NA + SUTVA
- But the *spread* of $\hat{\beta}$ across hypothetical samples — the standard error — depends on the error structure
- Wrong SE $\rightarrow$ wrong t -statistic $\rightarrow$ wrong p -value $\rightarrow$ wrong conclusion

Two problems: heteroskedasticity, and within-cluster (= serial) correlation.

What an SE *is*: the variance of $\hat{\beta}$ across hypothetical samples

- $\hat{\beta}$ is a **random variable**. Different samples from the same DGP give different $\hat{\beta}$'s.
- The **sampling distribution** of $\hat{\beta}$ is the distribution we'd see if we re-drew the sample many times.
- The **standard error** is the standard deviation of that distribution: $SE(\hat{\beta}) = \sqrt{\text{Var}(\hat{\beta})}$.

The weird part

We only ever see *one sample*. But we're trying to describe how $\hat{\beta}$ would *vary* across samples we never drew.

The SE formula is what lets us estimate the across-sample variance from a single sample.

The rest of this lesson is about whether we did that right.

Three flavors of standard error

- **Vanilla (homoskedastic)**: assumes $\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$ for all i
- **Robust (Eicker-Huber-White)**: lets each residual speak for itself — handles heteroskedasticity
- **Cluster-robust (CRVE)**: groups residuals by cluster, lets within-cluster errors correlate freely

Same point estimate, three different SEs. The SE you pick is a claim about the error structure.

The general formula: a “sandwich”

The variance of $\hat{\beta}$ across hypothetical samples has one general form:

$$\text{Var}(\hat{\beta}) = \underbrace{(X'X)^{-1}}_{\text{bread}} \underbrace{X' \text{Var}(\varepsilon | X) X}_{\text{meat}} \underbrace{(X'X)^{-1}}_{\text{bread}}.$$

- The **bread** $(X'X)^{-1}$ comes from OLS mechanics. We always have it.
- The **meat** $X' \text{Var}(\varepsilon | X) X$ is where the *error structure* lives.
- All three flavors (vanilla, robust, cluster-robust) have the same **bread**.
They differ only in how they estimate the meat.

The rest of this lesson: what goes in the middle, and what changes if we get it wrong.

Homoskedasticity: the sandwich collapses

Assumption. $\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$ (constant, doesn't depend on X_i).

Start from the general sandwich:

$$\text{Var}(\hat{\beta}) = (X'X)^{-1} (X' \text{Var}(\varepsilon \mid X) X) (X'X)^{-1}$$

Under homoskedasticity $\text{Var}(\varepsilon \mid X) = \sigma^2 \mathbf{I}$, so the **meat** becomes $\sigma^2(X'X)$, and the sandwich collapses:

$$\text{Var}(\hat{\beta}) = (X'X)^{-1} \sigma^2(X'X) (X'X)^{-1} = \sigma^2 (X'X)^{-1}.$$

- One scalar $\hat{\sigma}^2 = \frac{1}{n-k} \sum_i \hat{e}_i^2$ summarizes *all* the noise.
- This is what `lm()` reports by default. **Clean — only if homoskedasticity holds.**

Heteroskedasticity: the collapse breaks

Now $\sigma_i^2 = \text{Var}(\varepsilon_i \mid X_i)$ varies with i . The **meat** is:

$$X' \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2) X \neq \sigma^2(X'X).$$

- Each σ_i^2 on the diagonal is the variance of Y_i given X_i — the noise around the regression line for that observation.
- In a DiD with binary D , that means: treated observations have one variance, controls have another.
- No single σ^2 can scale this matrix out. **The collapse fails.**

Next: the Eicker–Huber–White fix.

Eicker–Huber–White: let each residual speak for itself

We don't know σ_i^2 . Replace it with the empirical $\hat{e}_i^2$. The sandwich stays a sandwich:

$$\widehat{\text{Var}}_{\text{HC}}(\hat{\beta}) = (X'X)^{-1} \left(\sum_i \hat{e}_i^2 X_i X_i' \right) (X'X)^{-1}.$$

- Each $\hat{e}_i^2$ enters *weighted* by that observation's leverage $X_i X_i'$.
- No assumption of constant variance. Each obs's noise speaks for itself.

The HC family — four small-sample corrections

HC0 (raw): the formula above, exactly. **HC1**: multiply by $n/(n - k)$ to correct for the degrees of freedom used fitting $\hat{\beta}$. Stata's default.

HC2 / HC3: also divide each $\hat{e}_i^2$ by $(1 - h_{ii})$ or $(1 - h_{ii})^2$ to penalize high-leverage observations.

HC1 is what we'll use below.

HC1 numerically: each $\hat{e}_i^2$ weighted by leverage

Six observations. Three pairs at $\tilde{x} \in \{-2, 0, +2\}$. Fitted $\hat{\beta}_1 = 1$; residuals shown.

i	cluster g	$\tilde{x}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$\tilde{x}_i^2 \cdot \hat{e}_i^2$
1	1	-2	+0.5	0.25	1.00
2	1	-2	+0.5	0.25	1.00
3	2	0	-1.0	1.00	0.00
4	2	0	-1.0	1.00	0.00
5	3	+2	+0.5	0.25	1.00
6	3	+2	+0.5	0.25	1.00
meat =					4.00

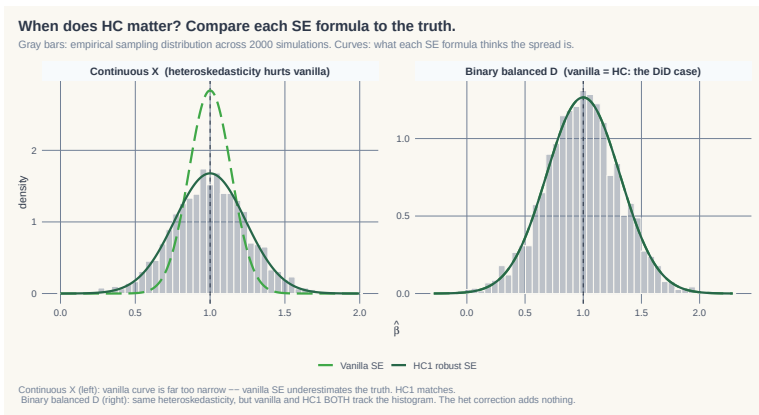
With $n = 6$, $k = 2$, and $S_{xx} = \sum_i \tilde{x}_i^2 = (-2)^2 + (-2)^2 + 0 + 0 + 2^2 + 2^2 = 16$:

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}_1) = \frac{n}{n-k} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} \cdot \underbrace{\text{meat}}_{4.00} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} = \frac{6}{4} \cdot \frac{4.00}{16 \cdot 16} = \frac{6}{4} \cdot \frac{4.00}{256} = 0.02344$$

$$\Rightarrow \text{SE}_{\text{HC1}} = \sqrt{0.02344} \approx 0.153$$

Obs 3–4 contribute zero because $\tilde{x} = 0$ kills the leverage weight. **HC1 treats all 6 obs as independent.**
 Scalar form shown gives $\text{SE}(\hat{\beta}_1)$ only (intercept absorbed via FWL). The matrix sandwich returns every coefficient's SE in one shot.

When does HC matter? Compare each SE formula to the truth.



Histogram = the true sampling distribution of $\hat{\beta}$ (2000 simulations). Curves = what each SE formula thinks.

Continuous X: vanilla curve is too narrow — vanilla underestimates SD by 44%; HC1 matches. **Binary balanced D:** vanilla and HC1 both match the histogram. The het correction adds nothing.

Why HC-robust barely helps in DiD

The HC correction reweights residuals by $X_i X_i'$. In a DiD with binary treatment $D_i \in \{0, 1\}$ and roughly balanced groups:

- Treated and control units have similar *leverage*: each side carries $\approx$ half the weight in $X'X$.
- Residual variance is roughly *symmetric* across the two groups (binary D doesn't generate a fan shape in $\hat{e}^2$).
- So the weighted sum $\sum_i \hat{e}_i^2 X_i X_i'$ ends up not very different from the unweighted $\hat{\sigma}^2(X'X)$.

Consequence

$SE_{HC} \approx SE_{vanilla}$ for typical DiD specifications.

The over-rejection problem is not heteroskedasticity. It's something else.

Bertrand, Duflo & Mullainathan (QJE 2004) — the placebo-law test

“How Much Should We Trust Differences-in-Differences Estimates?”

- BDM generate *placebo laws*: assign random treatment dates to states in a real CPS panel.
- True effect by construction: $\beta = 0$. So $\Pr(\text{reject } H_0)$ should equal $\alpha = 5\%$.
- **Vanilla SE**: rejection rate $\approx 45\%$. Wildly over-rejects.
- **Heteroskedasticity-robust (HC1)**: rejection rate *also* $\approx 45\%$. **HC barely moves the needle.**
- **Cluster-robust at state level**: rejection rate $\approx 5\%$. The fix.

Why HC doesn't save you in DiD

Binary, balanced D + within-state serial correlation. The het-correction has nothing to do with the actual problem. **Clustering is doing all the work.**

Cluster-robust SE: sum scores within cluster, *then* square

For each cluster g , the “score” is $s_g = \sum_{i \in g} \tilde{x}_i \hat{e}_i$.

$$\widehat{\text{Var}}_{\text{CRVE}}(\hat{\beta}) \propto (X'X)^{-1} \left(\sum_g s_g s_g' \right) (X'X)^{-1}$$

- HC1 squares each *observation's* contribution, then sums.
- CRVE *sums within a cluster first* — then squares. Cross-terms between same-cluster obs get added in.
- If errors are correlated within a cluster, those cross-terms are positive and the SE grows.

CRVE numerically: same 6 obs, group scores, then square

Same 6 observations. Now compute $s_g = \sum_{i \in g} \tilde{x}_i \hat{e}_i$ for each cluster, then s_g^2 :

i	g	$\tilde{x}_i$	$\hat{e}_i$	$\tilde{x}_i \hat{e}_i$			
					g	s_g	s_g^2
1	1	-2	+0.5	-1			
2	1	-2	+0.5	-1	1	-2	4
3	2	0	-1.0	0	2	0	0
4	2	0	-1.0	0	3	+2	4
5	3	+2	+0.5	+1	meat =		8
6	3	+2	+0.5	+1			

With $G = 3$, $n = 6$, $k = 2$, and $S_{xx} = \sum_i \tilde{x}_i^2 = (-2)^2 + (-2)^2 + 0 + 0 + 2^2 + 2^2 = 16$, so $S_{xx}^2 = 256$:

$$\widehat{\text{Var}}_{\text{CRVE}}(\hat{\beta}_1) = \frac{G}{G-1} \cdot \frac{n-1}{n-k} \cdot \frac{\text{meat}}{S_{xx}^2} = \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{8}{256} = 0.05859 \Rightarrow \text{SE}_{\text{CRVE}} \approx 0.242$$

HC1: 0.153 vs. **CRVE:** 0.242 (**58% larger**). Same data. The cluster cross-terms changed the **meat** from 4 to 8.

Two formulas side by side

HC1 (obs independent)

$$\frac{n}{n-k} \cdot \frac{\sum_i \tilde{x}_i^2 \hat{e}_i^2}{\left(\sum_i \tilde{x}_i^2\right)^2}$$

sum each $\hat{e}_i^2 \tilde{x}_i^2$ separately

CRVE (cluster-robust)

$$\frac{G}{G-1} \cdot \frac{n-k}{n-k} \cdot \frac{\sum_g s_g^2}{\left(\sum_i \tilde{x}_i^2\right)^2}$$

$s_g = \sum_{i \in g} \tilde{x}_i \hat{e}_i$, then sum s_g^2

- HC1 sums squared per-obs contributions: cross-terms within a cluster are *ignored*.
- CRVE sums within a cluster *before* squaring: cross-terms are *captured*.
- Matrix form: same sandwich $(X'X)^{-1}(\text{meat})(X'X)^{-1}$, but the **meat** differs.

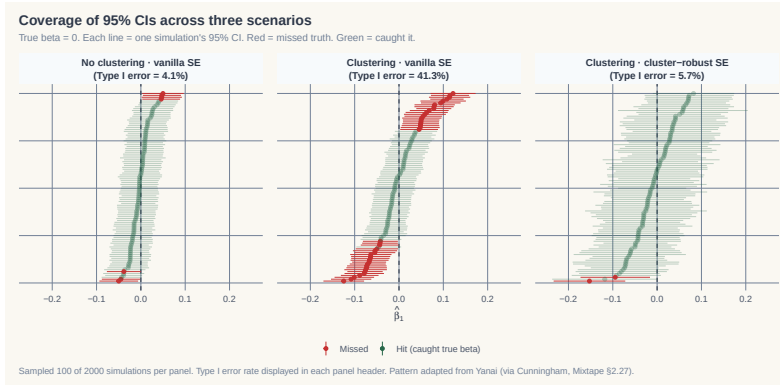
Same shape — *bread*, *meat*, *bread*. Difference: whether the **meat** sums over obs or over clusters.

The simulation we'll show: 2000 placebo regressions

- DGP: $y_{ig} = 0.4 + 0 \cdot x_{ig} + \varepsilon_{ig}$, 1000 obs, 50 clusters of 20.
- **True** $\beta_1 = 0$. Any rejection at $\alpha = 5\%$ is a false positive.
- Sweep three scenarios:
 - 1. No clustering ($\rho = 0$), vanilla SE
 - 2. Clustering ($\rho = 0.5$), vanilla SE
 - 3. Clustering ($\rho = 0.5$), *cluster-robust* SE
- For each scenario, draw 100 of the 2000 95% CIs and color them **Hit** or **Missed** against the true $\beta_1 = 0$.

*Pattern: Yanai $\rightarrow$ Cunningham, Mixtape §2.27. Script:
decks/code/SE-Demo/ci_forest_plot.R.*

Coverage of 95% CIs across the three scenarios



Left: no clustering, vanilla SE. Type I error $\approx 4\%$. As advertised.
Middle: clustering present, vanilla SE. **Type I error $\approx 41\%$.** Vanilla collapses.
Right: clustering, cluster-robust SE. **Back to $\approx 6\%$.** The fix works.

Drive the live app: two sliders, two stories

Open in your browser (runs entirely client-side via shinylive)

scunning.github.io/se-demo/

Two DGP knobs — ρ (clustering) and *het* (heteroskedasticity). Try these four corners:

ρ	<i>het</i>	What you see
0	0	All three SEs correct ($\approx 5\%$). Baseline.
0	high	Vanilla $\approx$ HC3 — balanced binary D makes het symmetric.
high	0	Vanilla $\approx$ HC3, both over-reject. CRVE is the fix.
high	high	Same. CRVE still the fix.

The lesson: with binary balanced treatment, *clustering* is the binding constraint.

LESSON

5. The Minimum Wage Wars

New Jersey raises its minimum wage. Pennsylvania doesn't.

- **April 1992**: New Jersey raises its minimum wage from \$4.25 to \$5.05 — a 19% increase.
- Pennsylvania, just over the river, stays at \$4.25.
- David Card and Alan Krueger survey 410 fast-food restaurants on both sides of the border, before (Feb 1992) and after (Nov 1992) the hike.
- The natural 2×2 : **treated = NJ, control = PA, pre/post the policy.**

Published in American Economic Review, 1994.

What economists *expected*

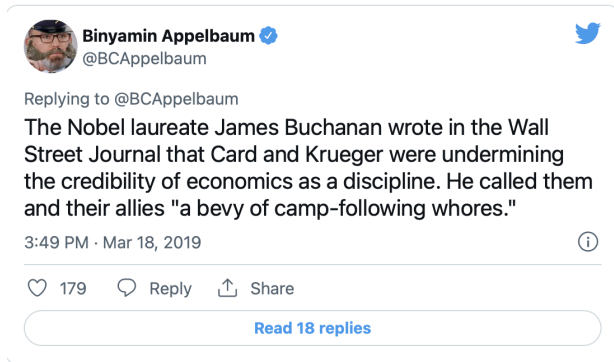
- Textbook supply and demand: a binding price floor above the market wage *must* reduce employment.
- Every undergraduate labor economics course taught this. Every PhD prelim tested it.
- Hundreds of empirical papers — mostly time-series, mostly cross-sectional — had found small *negative* employment effects of the minimum wage.
- Consensus among labor economists in 1992: **minimum wages cost jobs.**

What Card and Krueger *found*

Employment per restaurant	Pre (Feb 1992)	Post (Nov 1992)	Change
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	−2.16
DiD (NJ – PA)			+2.76 jobs/restaurant

- Employment in NJ fast food *rose* slightly, relative to PA, after the wage hike.
- Standard error puts the effect roughly indistinguishable from zero — but the sign is the opposite of what theory predicted.
- **No measurable disemployment effect.** The textbook had failed a real test.

The reaction was personal — Buchanan in the *Wall Street Journal*



James Buchanan (Nobel 1986), WSJ, April 25, 1996. Card and Krueger never named, but obvious.
"No self-respecting economist would claim that increases in the minimum wage increase employment. Such a claim . . . becomes equivalent to a denial that there is even minimal scientific content in economics."

Card on what publishing the paper cost him

I've subsequently stayed away from the minimum wage literature for a number of reasons. First, it cost me a lot of friends. People that I had known for many years . . . became very angry or disappointed. They thought that in publishing our work we were being traitors to the cause of economics as a whole.

— David Card, interview, after the publication

The price of a credible empirical paper that contradicted the consensus: years of professional and personal cost. Krueger paid more.

Hear it from them — three short interview clips

- **Orley Ashenfelter on the profession's reaction** (start 31:20)
youtu.be/M0tbuRX4eyQ?t=1882
- **Card on the hostility from peers** (1:07–1:10)
youtu.be/1soLdywFb_Q?t=4020
- **Card on what happened to Alan Krueger** (1:16–1:17)
youtu.be/1soLdywFb_Q?t=4556

The DiD literature did not arrive politely. It was earned through paper-by-paper combat, much of it personal.

Neumark and Wascher push back — and the fight continues

- Neumark & Wascher (2000): different data — BLS administrative payroll records instead of phone surveys.
- Found a small *negative* employment effect: -4% to -5% .
- Card & Krueger (2000) counter-replicate with their own administrative data. Still no effect.
- Three decades later, with hundreds of papers and dozens of states, the field has shifted: meta-analyses (Cengiz et al. 2019, Dube et al.) find **employment effects close to zero** at typical wage hike magnitudes.

What survived all of this

The point estimate was contested. **The 2×2 design framework was not.** The fight made DiD the standard — every minimum-wage paper since uses it.

We'll return to the minimum wage — with AI agents

- Card-Krueger's design changed labor economics. The DiD it used — one cohort, one switch date, NJ vs. PA — was the easy case.
- Modern minimum-wage research uses **staggered adoption**: many states, raising at different times, often by different amounts. The 2×2 stops being enough.
- Day 2 covers the staggered-adoption fix — Goodman-Bacon, **Callaway-Sant'Anna**, Sun-Abraham, BJS, dCdH.

Tomorrow: my new study using AI agents

I have a paper in progress using **LLM agents to study minimum-wage DiD** — asking the design questions Card and Krueger had to fight for paper by paper. I'll walk through it on Day 2 once we have CS on the board.

The design lesson: pre-trends are the gut check

- Card-Krueger held up because in the *pre-period*, NJ and PA fast-food employment moved together.
- If NJ employment had already been diverging before April 1992, the +2.76 estimate would have been mostly trend, not policy.
- Flat pre-trends → evidence (not proof) the design holds.
- **The 2×2 was credible because of the pre-period parallelism — not because of the regression coefficient.**

More tomorrow: when pre-trends look flat but the test still over-rejects (Roth, Roth-Sant'Anna).

LESSON

6. Event Studies

as Falsifications — and 2×2 s

DiD is a courtroom case

Scientific theories achieve credibility not just by explaining observed phenomena, but by excluding alternative explanations.

— Scott Cunningham, *The Remix* (forthcoming, Yale 2026)

- You are the prosecutor. The DiD coefficient is your **claim of guilt**.
- A claim is not evidence. The evidence is everything you do around the claim.
- Popper: a theory earns credibility by deriving what *e*/se should be true if it's right — and testing those predictions.

Five parts of a strong DiD

1. **Bite** — the policy actually changed what it was supposed to change
2. **Event study** — pre/post coefficients around treatment
3. **Falsification** — run a rival theory on a placebo group or outcome and try to reject *it*
4. **Main result** — the DiD on the actual outcome
5. **Mechanism** — the causal pathway from D to Y

The point estimate alone is a case “so weak that readers and the public are justified to reject it.”

A falsification puts a rival theory on trial

Write down what *else* the rival theory would predict — something irrelevant to your treatment — and test it.

- **Same outcome, alternative groups.** A group your treatment shouldn't reach, but the rival would.
Min wage on low-wage employment? Re-run on higher-wage workers.
- **Same group, alternative outcomes.** Outcomes your treatment shouldn't move, but the rival would.
Cigarette tax on smoking-related hospitalizations? Re-run on appendicitis and fractures.

Failing to reject the rival's prediction weakens the rival — it never confirms you. That's the Popperian asymmetry.

A clean falsification: peer effects on height

- Christakis & Fowler (2007): peer-effect contagion in obesity
- Cohen-Cole & Fletcher (2008): re-run the same model on **height, acne, and headaches**
- All three show “contagion”

We have not been able to become taller by having tall friends.

— Cohen-Cole & Fletcher, paraphrased

The result isn't necessarily wrong. But the empirical model is suspect.

The event study is the falsification built into the timeline

- Every coefficient at $\tau = -2, -3, \dots, -q$ asks the same question:
Is something already moving the treated group relative to controls, before D turned on?
- That's a rival theory. The pre-period coefficients are its placebo test.
- This one is built into the regression. The others (placebo group, placebo outcome) you have to construct.

None of these falsifications is the whole case. Event study, placebo group, placebo outcome — each rules out a different rival theory. Together they are powerful.

An event study is the 2×2 viewed in motion

- Instead of one pre-period and one post-period, use *many*
- Estimate a separate effect at each event-time relative to treatment
- Plot them as a sequence: $\hat{\beta}_{-4}, \hat{\beta}_{-3}, \dots, \hat{\beta}_{-1}, \hat{\beta}_0, \hat{\beta}_{+1}, \dots$
- The graph tells two stories: **pre-trends** (left of zero) and **dynamics** (right of zero)

Every coefficient is a 2×2 against $t = -1$

The saturated event-study regression:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}\{\text{event time}_{it} = k\} \cdot D_i + \varepsilon_{it}$$

Drop $k = -1$ as the reference.

Every other $\hat{\beta}_k$ is then a 2×2 DiD between event-time k and the period just before treatment. Let's prove it numerically.

A tiny panel where you can do the math in your head

Cell means $\bar{Y}_{g,t}$. Two treated units (T), two controls (C). Treatment turns on at event-time 0.

Event time k	-3	-2	-1	0	+1
Treated $\bar{Y}_k^T$	7	8	9	13	16
Control $\bar{Y}_k^C$	2	3	4	5	6
$\bar{Y}_k^T - \bar{Y}_k^C$	5	5	5	8	10

- Pre-period: T-C gap is constant at 5. **Parallel trends.**
- True $ATT(0) = 3$, True $ATT(+1) = 5$.

Each $\hat{\beta}_k$ by hand — one formula, four answers

The 2×2 against $t = -1$: $\hat{\beta}_k = (\bar{Y}_k^T - \bar{Y}_{-1}^T) - (\bar{Y}_k^C - \bar{Y}_{-1}^C)$

k	Calculation	$\hat{\beta}_k$
-3	$(7 - 9) - (2 - 4) = -2 - (-2)$	0
-2	$(8 - 9) - (3 - 4) = -1 - (-1)$	0
0	$(13 - 9) - (5 - 4) = 4 - 1$	3
+1	$(16 - 9) - (6 - 4) = 7 - 2$	5

Same arithmetic each row. Pre $\rightarrow$ zeros (PT holds). Post $\rightarrow$ the true ATT.

Same numbers via OLS in R

```
library(fixest)
feols(Y ~ i(event_time, treat, ref = -1) | unit + period,
      data = toy)
```

coefficient	estimate
event_time::-3:treat	0
event_time::-2:treat	0
event_time::0:treat	3
event_time::1:treat	5

OLS matches the manual 2×2 s to machine epsilon. The regression is the formula.

You could have picked a different hinge

Re-compute $\hat{\beta}_{-3}$, but with $t = -2$ as the baseline:

$$\tilde{\beta}_{-3} = (\bar{Y}_{-3}^T - \bar{Y}_{-2}^T) - (\bar{Y}_{-3}^C - \bar{Y}_{-2}^C) = (7 - 8) - (2 - 3) = \mathbf{0}$$

And the post-period one with $t = -2$ as the baseline:

$$\tilde{\beta}_0 = (\bar{Y}_0^T - \bar{Y}_{-2}^T) - (\bar{Y}_0^C - \bar{Y}_{-2}^C) = (13 - 8) - (5 - 3) = \mathbf{3}$$

What OLS *can't* give you simultaneously

A single regression drops *one* event-time dummy. Multicollinearity won't let it report 2×2 s against both $t = -1$ and $t = -2$ at once. The convention is $t = -1$. It's a choice, not math.

Why the hinge matters for the pre-trends test

The pre-period coefficients $\hat{\beta}_{-q}, \dots, \hat{\beta}_{-2}$ are:

$$\hat{\beta}_k = \underbrace{(\bar{Y}_k^T - \bar{Y}_{-1}^T)}_{\text{long-difference, treated}} - \underbrace{(\bar{Y}_k^C - \bar{Y}_{-1}^C)}_{\text{long-difference, control}}$$

- These are *long-difference trends from $t = -1$* — not point-to-point changes.
- A **pre-trends check** asks whether those long differences are statistically zero.
- Post-period coefficients use the *same hinge*. The pre-period model has to match the long-difference structure of the post-period model.

Match the model, match the test. A different baseline is a different question.

Pre-trends are not parallel trends

The **parallel-trends assumption** (post-period, counterfactual):

$$\{ \mathbb{E}[Y(0)|D=1, t] - \mathbb{E}[Y(0)|D=1, t-1] \} - \{ \mathbb{E}[Y(0)|D=0, t] - \mathbb{E}[Y(0)|D=0, t-1] \} = 0$$

Parallel pre-trends (pre-period, observable):

$$\{ \mathbb{E}[Y(0)|D=1, t-2] - \mathbb{E}[Y(0)|D=1, t-1] \} - \{ \mathbb{E}[Y(0)|D=0, t-2] - \mathbb{E}[Y(0)|D=0, t-1] \} = 0$$

- Different periods. Different conditional expectations. **Different quantities.**
- Neither implies the other. Pre-trends can be flat and post-period PT can fail. Pre-trends can wobble and post-period PT can still hold.
- Flat pre-trends are **heuristic evidence**, not a test of the identifying assumption.

castle.dta event study, live

```
library(fixest); library(ggplot2)
df <- read_dta("castle.dta")
df <- df %>% mutate(event_time = year - effyear)

es <- feols(l_homicide ~ i(event_time, ref = -1) | sid + year,
            data = df, cluster = ~sid)

ipplot(es, main = "Event study: castle.dta")
```

Full script: CI-2/Lab/Basic-DiD/simple_eventstudy.R

LESSON

7. Two ATTs, Two PTs

A callback to weighting

There isn't *one* parallel trends

Remember Lesson 3: per-person ATE and per-county ATE were different parameters under selection. Sometimes different signs.

The 2×2 decomposition has *two* terms

$$\hat{\delta} = ATT + PT$$

Weighting changes the expectation operator. So *both* terms get re-weighted.

- Weighted vs unweighted DiD: different ATTs. We've seen this.
- Weighted vs unweighted DiD: **different parallel-trends assumptions**. That's new.
- Two parameters. Two identifying assumptions. *Not* the same 2×2 .

Two decompositions, four assumptions

	Unweighted 2×2 <i>average county</i>	Population-weighted 2×2 <i>average person</i>
Estimator	$\hat{\delta}_{\text{unw}}$	$\hat{\delta}_{\text{wtd}}$
Target parameter	ATT_{unw}	ATT_{wtd}
PT assumption	$\mathbb{E}[\Delta Y(0) \mid D = 1] = \mathbb{E}[\Delta Y(0) \mid D = 0]$	$\tilde{\mathbb{E}}[\Delta Y(0) \mid D = 1] = \tilde{\mathbb{E}}[\Delta Y(0) \mid D = 0]$

- $\tilde{\mathbb{E}}[\cdot]$ is the population-weighted expectation (each county weighted by its size S).
- **Same data. Two estimators. Two different parameters. Two different PT assumptions.**

First, what *is* the weighted mean?

For any variable Y and any weight S :

$$\tilde{\mathbb{E}}[Y] \equiv \frac{\mathbb{E}[S Y]}{\mathbb{E}[S]}.$$

- Each unit's Y is multiplied by its weight S . Bigger units count more.
- Divide by total weight $\mathbb{E}[S]$ to recover an average.

What we're about to prove

$$\tilde{\mathbb{E}}[Y] = \mathbb{E}[Y] + \frac{\text{Cov}(S, Y)}{\mathbb{E}[S]}$$

Proving it: five lines from first principles

1. **Definition.** $\tilde{\mathbb{E}}[Y] = \frac{\mathbb{E}[S Y]}{\mathbb{E}[S]}$
2. **Covariance.** $\text{Cov}(S, Y) = \mathbb{E}[S Y] - \mathbb{E}[S] \mathbb{E}[Y]$
3. **Solve for $\mathbb{E}[S Y]$.** $\mathbb{E}[S Y] = \mathbb{E}[S] \mathbb{E}[Y] + \text{Cov}(S, Y)$
4. **Substitute into line 1.** $\tilde{\mathbb{E}}[Y] = \frac{\mathbb{E}[S] \mathbb{E}[Y] + \text{Cov}(S, Y)}{\mathbb{E}[S]}$
5. **Distribute.** $\tilde{\mathbb{E}}[Y] = \mathbb{E}[Y] + \frac{\text{Cov}(S, Y)}{\mathbb{E}[S]} \quad \blacksquare$

Two definitions and three algebraic moves. That's the entire identity.

What the identity says

$$\tilde{\mathbb{E}}[Y] = \mathbb{E}[Y] + \frac{\text{Cov}(S, Y)}{\mathbb{E}[S]}$$

- Weighted mean = unweighted mean + a **covariance correction**.
- Correction is zero $\iff S$ is uncorrelated with Y . Otherwise the two means diverge.
- $\text{Cov}(S, Y) > 0$ (S co-moves with Y across units): the weighted mean exceeds the unweighted.
- $\text{Cov}(S, Y) < 0$ (S moves opposite Y): the weighted mean is smaller.

Next: apply this identity to $\Delta Y(0)$, conditional on D . Watch the PT term split into two pieces.

Apply to $\Delta Y(0)$, conditional on $D=d$

Substitute $\Delta Y(0)$ for Y , condition on $D=d$:

$$\tilde{\mathbb{E}}[\Delta Y(0) \mid D=d] = \mathbb{E}[\Delta Y(0) \mid D=d] + \underbrace{\frac{\text{Cov}(S, \Delta Y(0) \mid D=d)}{\mathbb{E}[S \mid D=d]}}_{\kappa_d}$$

Subtract the $D=0$ version from the $D=1$ version. Unweighted differences give PT_{unw} ; weighted ones give PT_{wtd} :

$$\text{PT}_{\text{wtd}} = \text{PT}_{\text{unw}} + (\kappa_1 - \kappa_0)$$

Reading κ_d : a covariance, scaled, *within* group d

$$\kappa_d = \frac{\text{Cov}(S, \Delta Y(0) \mid D=d)}{\mathbb{E}[S \mid D=d]}$$

- **Numerator:** within group d , does unit size S co-move with the untreated trend $\Delta Y(0)$? Positive $\Rightarrow$ big units trend up faster within d .
- **Denominator:** average size in group d . Normalizes so κ_d is in $\Delta Y(0)$ units.
- **Subscript:** computed *separately* for treated ($d=1$) and control ($d=0$).

The covariance gap

$$\kappa_1 - \kappa_0 = \text{“does the size–trend relationship differ across treated and control?”}$$

The knife-edge: both PTs hold iff $\kappa_1 = \kappa_0$

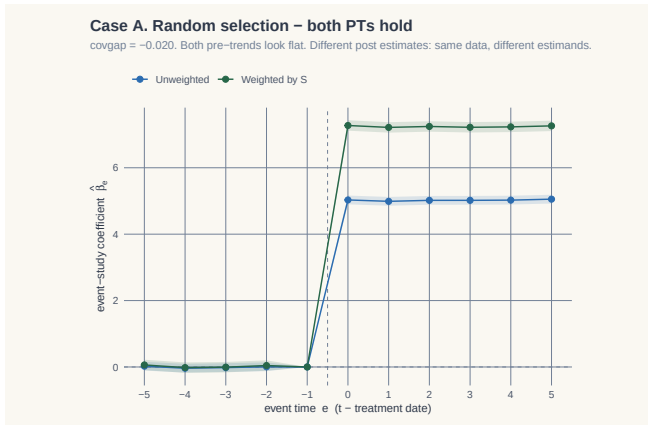
$$PT_{\text{wtd}} = PT_{\text{unw}} + (\kappa_1 - \kappa_0)$$

- $\kappa_1 = \kappa_0 = 0$ is *not* required. The covariances can be huge — they just have to be *equal* across treated and control.
- If $\kappa_1 \neq \kappa_0$: **at most one of the two PTs can be true.**
- One 2×2 design can identify ATT_{unw} , or ATT_{wtd} , but in general *not both*.

What creates a nonzero covariance gap?

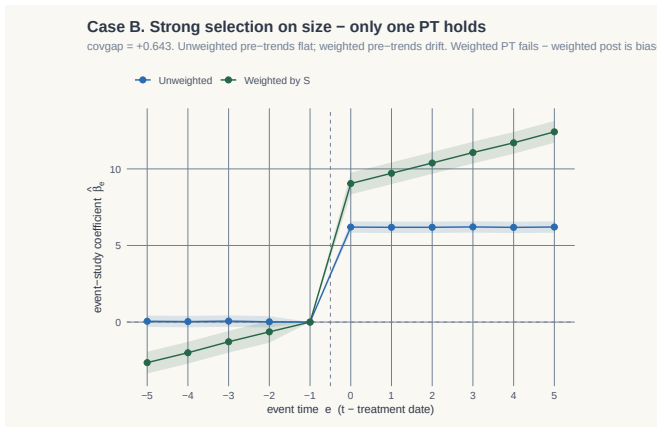
Selection on the weight. Big counties get treated for reasons *related to their size*, and big counties' $Y(0)$ trends differently than small ones'. The two together produce the gap.

Case A: covgap ≈ 0 — both pre-trends look flat



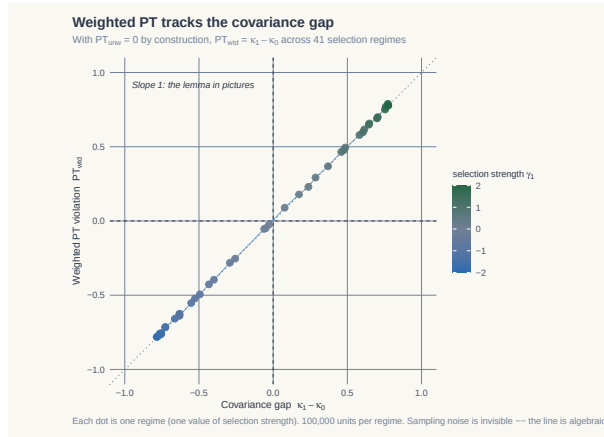
Same data. Both event studies look clean. **Different post estimates — because they target different ATTs.**

Case B: covgap large — the weighted event study breaks



Selection on size shifts $\kappa_1 - \kappa_0$ away from zero. The unweighted event study still passes; the weighted one drifts. **The pre-trend test belongs to the assumption you're running.**

Simulation: the weighted PT *is* the covariance gap



$N=100,000$, $PT_{unw} = 0$ by construction. Sweep selection strength γ_1 from -2 to $+2$.
 PT_{wtd} traces $\kappa_1 - \kappa_0$ exactly. Slope 1. Not a regularity — an identity.

Weighted vs unweighted is *not* a robustness check

- Running both and hoping they agree is *not* a robustness check — it's a comparison of two different estimands under two different assumptions.
- Disagreement is information about the **covariance gap**, not about whether DiD “works.”
- Pick the parameter your question demands. Then defend the corresponding PT.

So what do you do?

Event studies (we just saw) interrogate *which* PT you can defend. You can run them weighted, unweighted, or both — but each tests a different assumption.

Derivation and simulation: `decks/code/Weighted-PT/weighted_pt_chart.R`.

Background: Scott's Substack, “Diff-in-Diff, population weights and parallel trends,” Part 1 & 2.

A closing example: Medicaid expansion and mortality

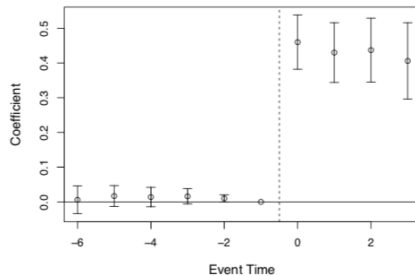
Miller, Johnson & Wherry (2021), *Quarterly Journal of Economics*.

- Did the ACA Medicaid expansion (2014) reduce mortality among the **near-elderly** (55–64)?
- Comparison: states that expanded vs. states that didn't, before vs. after 2014.
- Why *near-elderly*? Because everyone 65+ has Medicare — Medicaid expansion is irrelevant for them.
The elderly become a natural **placebo group** living in the same economies.

Walkthrough

We'll see the **bite** (three measures), the **falsification** (elderly placebo), and the **main mortality event study**. That's five figures. They build the case in sequence — not in parallel.

Bite #1: Medicaid take-up jumped in expansion states

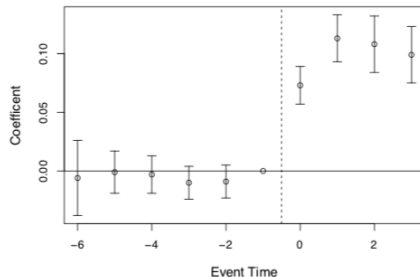


(a) Medicaid Eligibility

Did the policy actually move people onto Medicaid? Yes — enrollment rose sharply in expansion states post-2014, flat in non-expansion.

Without bite, there's no story. The mandate has to actually do something before we can attribute downstream changes to it.

Bite #2: any-insurance coverage rose

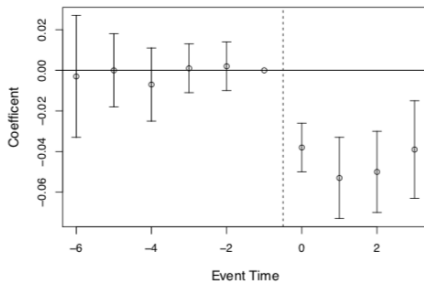


(b) Medicaid Coverage

Did the new Medicaid enrollees actually gain coverage they didn't have? Or did they just shift from private to Medicaid?

Any-insurance coverage rose — the expansion brought in the previously uninsured, not the already-insured.

Bite #3: the uninsured rate fell

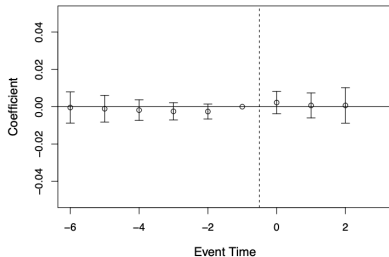


(c) Uninsured

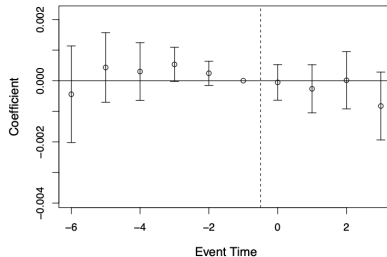
*The mirror image of Bite #2. Uninsured rate dropped sharply in expansion states post-2014. Three measures (Medicaid up, any-insurance up, uninsured down) converge on the same conclusion: **the policy bit.***

Falsification: same model, on the 65+ population — no effect

Age 65+ in 2014



(c) Medicaid Coverage



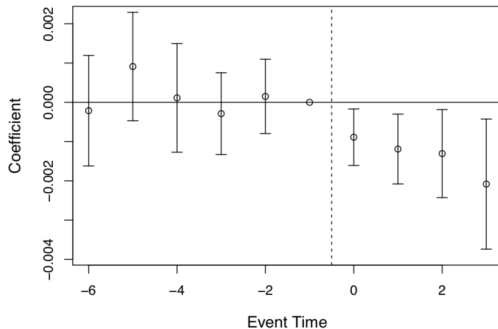
(d) Annual Mortality

Same states, same years, same model — but run on the 65+ population. *They all have Medicare, so Medicaid expansion shouldn't touch their mortality.*

A rival theory ("some 2014 shock hit expansion states") would predict a mortality effect here. It didn't appear.

Rival rejected.

The main result: $\approx 9.2\%$ reduction in near-elderly mortality



*Pre-trends flat. Post-2014 mortality declines in expansion states. $\approx 9.2\%$ **reduction**.
Effect grows over time — consistent with cumulative treatment for chronic conditions.*

The five parts of a strong DiD, all five present

Bite	Three figures: Medicaid take-up up, any-insurance up, uninsured down
Event study	Pre-trends flat; post-trends decline in expansion states
Falsification	No effect on elderly (65+) — same states, same years, same model
Main result	$\approx 9.2\%$ reduction in mortality among the near-elderly
Mechanism	Disease-related deaths fall and compound; consistent with treated chronic conditions

*None of these alone is the case. **Together they are powerful.***

What pre-trends do and don't tell us

- Flat pre-trends → evidence consistent with parallel trends
- Sloped pre-trends → evidence *against* parallel trends
- Flat pre-trends $\nRightarrow$ parallel trends *must* hold post-treatment
- It's one falsification. Run the others too — placebo groups, placebo outcomes.
- **Pre-trends are a gut check, not a proof.** More tomorrow (Roth-Sant'Anna, Ghanem-Sant'Anna-Wüthrich).

Try it yourself: the Event-Study Explorer

Open in your browser (runs entirely client-side via shinylive)

scunning1975.github.io/event-study/

Five sliders. Watch the event-study coefficients trace the truth in real time.

- Set the **true treatment effect** (level) and **effect slope**. Watch the post-period coefs trace the staircase.
- Add a **pre-trend slope**. **Watch the pre-period coefs slope away from zero — the failed gut-check, visible.**
- Crank up the **noise** or shrink the **number of units**. Watch the CIs blow out.

Source: `decks/code/Event-Study-Demo/app.R`. Single-cohort DiD with TWFE event study. Day 2 introduces the staggered version.

What we've done today

1. Three founders, one design — four averages, three subtractions
2. Three regressions, one answer
3. Potential outcomes, target parameters, and parallel trends
4. Standard errors: when robust isn't robust enough
5. The minimum wage wars
6. Event studies as falsifications and 2×2 s
7. Two ATTs, two parallel trends — a callback to weighting

Tomorrow: when the 2×2 stops working

- Covariates — IPW, OR, doubly robust
- Time-varying covariates (Caetano & Callaway)
- Triple differences (Ortiz-Villavicencio & Sant'Anna)
- Staggered adoption — Goodman-Bacon, Callaway-Sant'Anna, Sun-Abraham, BJS

DiD is a research design
with two ingredients —
the **2×2 calculation** and **parallel trends**.
Together they identify the ATT.